**MTH109**
**Specimen Examination Paper**
**Guided Solutions**

## Question 1

Determine the following limits.

(a) $\lim_{x\to 1} \frac{\sqrt{x^2-2x+5}}{x^3+5}$

(5 marks)

(b) $\lim_{x\to \infty} \frac{\sqrt{25x^6-3}}{3x^3-2x^2+4x+1}$

(6 marks)

(c) $\lim_{x\to 0^+} x^{x^{\tan x}}$

(7 marks)

(d) $\lim_{x\to -\infty} \frac{x^3(2\sin x+3\cos^4 x)}{x^4+1}$

(7 marks)

**<u>Marking Guide:</u>**

(a) By the root law for limits and the limit laws, as well as the theorem on evaluating limits of rational functions, we can calculate the limit by substituting $x = 1$ into the expression, giving

$$\lim_{x\to 1} \frac{\sqrt{x^2-2x+5}}{x^3+5} = \frac{\sqrt{1^2-2(1)+5}}{1^3+5} = \frac{2}{6} = \frac{1}{3}.$$

(5 marks)

(b) $\lim_{x\to \infty} \frac{\sqrt{25x^6-3}}{3x^3-2x^2+4x+1} = \lim_{x\to \infty} \frac{\frac{1}{x^3}\sqrt{25x^6-3}}{\frac{1}{x^3}(3x^3-2x^2+4x+1)} = \lim_{x\to \infty} \frac{\sqrt{25-\frac{3}{x^6}}}{3-\frac{2}{x}+\frac{4}{x^2}+\frac{1}{x^3}} = \frac{\sqrt{25}}{3} = \frac{5}{3}.$

(6 marks)

(c) Let $y = x^{x^{\tan x}}$. Then,

$$\log y = x^{\tan x} \log x.$$

(2 marks)

Observe that $x^{\tan x} = e^{\tan x \log x}$, and using L'Hôpital's rule gives

$$\lim_{x\to 0^+} \tan x \log x = \lim_{x\to 0^+} \frac{\log x}{\cot x} = \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\csc^2 x} = \lim_{x\to 0^+} -\frac{\sin x}{x} \cdot \sin x = -1 \cdot 0 = 0.$$

(3 marks)

Therefore

$$\lim_{x\to 0^+} x^{\tan x} = e^0 = 1.$$

Thus,

$$\lim_{x\to 0^+} \log y = \lim_{x\to 0^+} x^{\tan x} \log x = -\infty,$$

which gives

$$\lim_{x\to 0^+} y = \lim_{t\to -\infty} e^t = 0.$$

(2 marks)
(Total: 7 marks)

(d) Since $-1 \le \sin x \le 1$ and $0 \le \cos^4 x \le 1$, one has

$$-2 \le 2\sin x + 3\cos^4 x \le 5.$$

(2 marks)

As we are computing the limit as $x \to -\infty$, it may be assumed that $\frac{x^3}{x^4+1} \le 0$, and therefore

$$\frac{-2x^3}{x^4+1} \ge \frac{x^3(2\sin x + 3\cos^4 x)}{x^4+1} \ge \frac{5x^3}{x^4+1}.$$

(2 marks)

Since $\lim_{x\to -\infty} \frac{-2x^3}{x^4+1} = \lim_{x\to -\infty} \frac{\frac{-2}{x}}{1+\frac{1}{x^4}} = 0$ and $\lim_{x\to -\infty} \frac{5x^3}{x^4+1} = \lim_{x\to -\infty} \frac{\frac{5}{x}}{1+\frac{1}{x^4}} = 0$, it follows from the Squeeze Theorem that

$$\lim_{x\to -\infty} \frac{x^3(2\sin x + 3\cos^4 x)}{x^4+1} = 0.$$

(3 marks)
(Total: 7 marks)

**Question 2**

(a) Apply derivative tests to find the absolute and local maxima and minima of the following function on $[-1,1]$.

$$f(x) = \begin{cases} -x^2 - x + \pi, & \text{if } -1 \le x \le 0; \\ \dfrac{sin(\pi x)}{x(1-x)}, & \text{if } 0 < x < 1; \\ \pi, & \text{if } x = 1. \end{cases}$$

(15 marks)

(b) Define the function $F: [0, \infty) \longrightarrow \mathbb{R}$ by $F(x) = \sqrt[4]{x}$. Use an appropriate Taylor polynomial to estimate $F(16.01)$ correct to 4 decimal places.

(10 marks)

**<u>Marking Guide:</u>**

(a) The function $f$ has domain $[0,1]$ and is composed of the continuous function $-x^2 - x + \pi$ on $[-1,0)$ and the function $\frac{sin(\pi x)}{x(1-x)}$ on $(0,1)$, where the latter function is continuous on $(0,1)$.
Since

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (-x^2 - x + \pi) = \pi$$

and, by L'Hôpital's rule,

$$lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \frac{sin(\pi x)}{x(1-x)} = \lim_{x \to 0^+} \frac{\pi\, cos(\pi x)}{1 - 2x} = \pi,$$

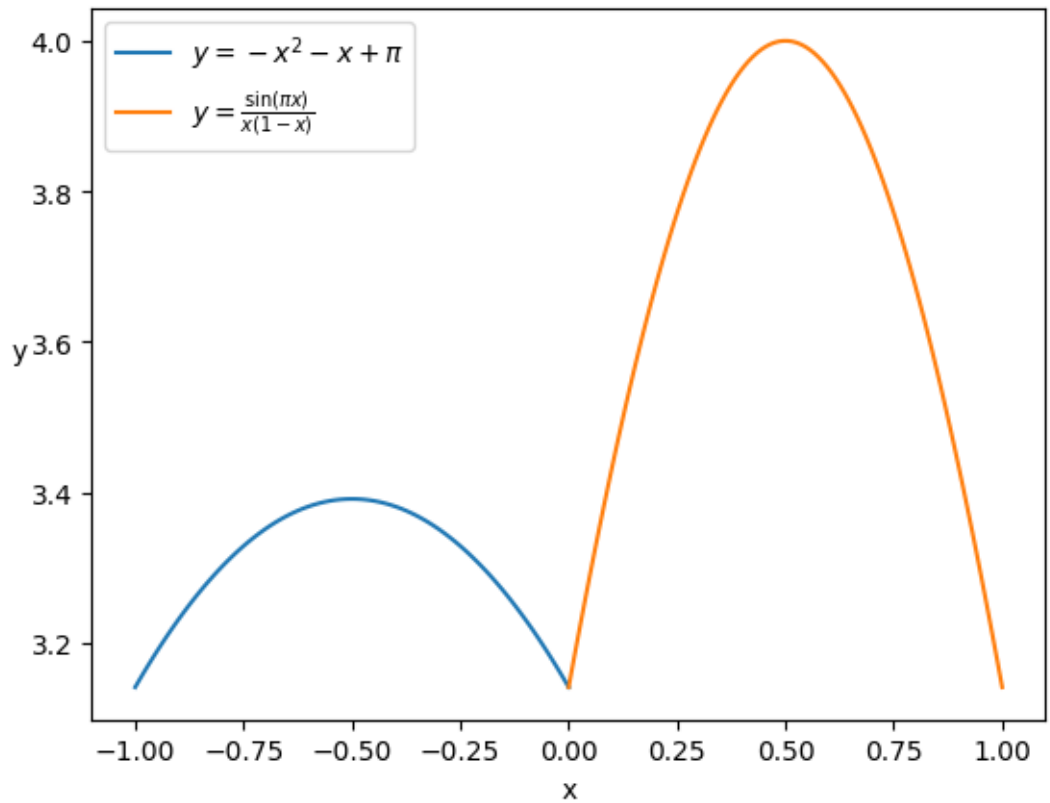
$f$ is continuous at $x = 0$. Similarly, L'Hôpital's rule gives

$$lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} \frac{sin(\pi x)}{x(1-x)} = \lim_{x \to 1^-} \frac{\pi\, cos(\pi x)}{1 - 2x} = \pi,$$

and since $f(1) = \pi$, it follows that $f$ is continuous at $x = 1$.

(2 marks)

The graph of $f$ is shown in the next figure.




When $-1 \leq x < 0$, $f'(x) = -2x - 1$. When $0 < x < 1$,

$$f'(x) = \frac{x(1-x)\pi\, cos(\pi x) - sin(\pi x)\,(1-2x)}{x^2(1-x)^2}.$$

We first determine the critical points of $f$. At $x = 0$,

$$\lim_{x\to 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0^-} (-2x - 1) = -1,$$

$$\lim_{x\to 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0^+} \frac{x(1-x)\pi\, cos(\pi x) - sin(\pi x)\,(1-2x)}{x^2(1-x)^2}$$

$$= \lim_{x\to 0^+} \frac{-\pi^2(x - x^2)\, sin(\pi x) + \pi(1-2x)\, cos(\pi x) - (-2)\, sin(\pi x) - \pi(1-2x)\, cos(\pi x)}{-2x^2(1-x) + 2x(1-x)^2}$$

$$= \lim_{x\to 0^+} \frac{sin(\pi x)\,(\pi^2 x^2 - \pi^2 x + 2)}{2x(1-x)(1-2x)} = \lim_{x\to 0^+} \frac{sin(\pi x)}{\pi x} \cdot \lim_{x\to 0^+} \frac{\pi}{2} \cdot \frac{\pi^2 x^2 - \pi^2 x + 2}{(1-x)(1-2x)} = 1 \cdot \frac{\pi}{2} \cdot 2$$

$$= \pi.$$

Hence $f$ is not differentiable at $x = 0$, so $x = 0$ is a critical point of $f$. Since $f'(x) = -2x - 1 < 0$ when $x$ is close to and less than 0 and $f'(x) = \frac{x(1-x)\pi\, cos(\pi x) - sin(\pi x)(1-2x)}{x^2(1-x)^2} > 0$ when $x$ is close to and more than 0, $f$ has a local minimum at $x = 0$. Next, we consider the points where $f'(x) = 0$ on the intervals $(-1,0)$ and $(0,1)$.

When $-2x - 1 = 0$, $x = -\frac{1}{2}$. Furthermore, $f''\left(-\frac{1}{2}\right) = -2 < 0$, and therefore $f$ has a local maximum at $x = -\frac{1}{2}$.

(2 marks)

Next, setting $\frac{x(1-x)\pi\, cos(\pi x) - sin(\pi x)(1-2x)}{x^2(1-x)^2} = 0$ gives the equation

$$g(x) = x(1-x)\pi\, cos(\pi x) - sin(\pi x)\,(1-2x) = 0.$$

Observe that the latter equation has the solutions $x = 0,1$ (at which $\frac{x(1-x)\pi\, cos(\pi x) - sin(\pi x)(1-2x)}{x^2(1-x)^2}$ is undefined) and $x = \frac{1}{2}$. Hence $f'\left(\frac{1}{2}\right) = 0$. Furthermore,

$$g'(x) = sin(\pi x)\,(\pi^2 x^2 - \pi^2 x + 2),$$

and for all $x \in (0,1)$, $g'(x) = 0$ if only if

$$\pi^2\left(\left(x - \frac{1}{2}\right)^2 + \frac{2}{\pi^2} - \frac{1}{4}\right) = 0,$$

or

$$x = \frac{1}{2} \pm \frac{\sqrt{\pi^2 - 8}}{2\pi}.$$

When $x \in \left(0, \frac{1}{2} - \frac{\sqrt{\pi^2-8}}{2\pi}\right)$ or $x \in \left(\frac{1}{2} + \frac{\sqrt{\pi^2-8}}{2\pi}, 1\right)$, $g'(x) > 0$, and when $x \in \left(\frac{1}{2} - \frac{\sqrt{\pi^2-8}}{2\pi}, \frac{1}{2} + \frac{\sqrt{\pi^2-8}}{2\pi}\right)$, $g'(x) < 0$. Hence, since $g\left(\frac{1}{2}\right) = 0$, there is some $\delta > 0$ such that $g(x) > 0$ (and hence $f'(x) > 0$) for each $x \in \left(\frac{1}{2} - \delta, \frac{1}{2}\right)$ and $g(x) < 0$ (and hence $f'(x) < 0$) for each $x \in \left(\frac{1}{2}, \frac{1}{2} + \delta\right)$. Thus, $f$ has a local maximum at $x = \frac{1}{2}$.

(3 marks)

Since $g(0) = 0 = g\left(\frac{1}{2}\right)$, it follows from Rolle's theorem that if there were a point $x_1 \in \left(0, \frac{1}{2}\right)$ such that $g(x_1) = 0$, then there would be at least two points $y_1 \in (0, x_1)$ and $y_2 \in \left(x_1, \frac{1}{2}\right)$ such that $g'(y_1) = 0$ and $g'(y_2) = 0$. However, as shown earlier, there is only one real root of $g'(x) = 0$ in the interval $\left(0, \frac{1}{2}\right)$. Hence the equation $g(x) = 0$ has no real root in $\left(0, \frac{1}{2}\right)$. Arguing similarly, since $g\left(\frac{1}{2}\right) = 0 = g(1)$, the equation $g(x) = 0$ has no real root in $\left(\frac{1}{2}, 1\right)$. Therefore $f$ has exactly one critical point, which occurs at $x = \frac{1}{2}$, over $(0,1)$.

The critical points of $f$ are therefore $x = -\frac{1}{2}$ (local maximum), $x = 0$ (local minimum) and $x = \frac{1}{2}$ (local maximum).

(5 marks)

To determine the absolute maxima and absolute minima of $f$, we also need to evaluate $f$ at the endpoints $x = -1$ and $x = 1$.

One has $f(-1) = \pi \approx 3.142$, $f\left(-\frac{1}{2}\right) = \pi + \frac{1}{4} \approx 3.392$, $f(0) = \pi \approx 3.142$, $f\left(\frac{1}{2}\right) = 4$ and $f(1) = \pi \approx 3.142$. Comparing these values, we conclude that over $[-1,1]$, $f$ has an absolute minimum value of $\pi$ at $x = -1, 0, 1$ and an absolute maximum value of 4 at $x = \frac{1}{2}$.

(3 marks)
(Total: 15 marks)

(b) We compute the first-order Taylor polynomial of $F$ at $a = 16$. First,

$$F(16) = \sqrt[4]{16} = 2,$$
$$F'(x) = \frac{1}{4}x^{-\frac{3}{4}}, F'(16) = \frac{1}{32},$$
$$F''(x) = -\frac{3}{16}x^{-\frac{7}{4}}.$$

(3 marks)

Hence the first-order Taylor polynomial of $F$ at $a = 16$ is

$$p_1(x) = F(16) + F'(16)(x - 16) = 2 + \frac{x - 16}{32}.$$

(3 marks)

At $x = 16.01$, the remainder term $R_1(16.01)$ is given by

$$R_1(16.01) = \frac{F''(c)}{2!}(16.01 - 16)^2 = -\frac{3}{32}c^{-\frac{7}{4}}(0.01)^2 = -\frac{3}{32} \times 10^{-4} \times c^{-\frac{7}{4}}$$

for some $c \in (16,16.01)$.

(2 marks)

Since $\frac{3}{32} \times 10^{-4} \times x^{-\frac{7}{4}}$ is a decreasing function of $x$ on $(0, \infty)$, it follows that

$$|R_1(16.01)| = \frac{3}{32} \times 10^{-4} \times c^{-\frac{7}{4}} < \frac{3}{32} \times 10^{-4} \times 16^{-\frac{7}{4}} < 8 \times 10^{-8} < 10^{-5},$$

and thus $p_1(16.01) = 2 + \frac{16.01-16}{32} = 2.0003125 \approx 2.0003$ approximates $F(16.01)$ up to an accuracy of 4 decimal places.

(2 marks)
(Total: 10 marks)

**Question 3**

Solve for and simplify the following derivative:

$$\frac{d}{dx}\int_{\cos x}^{\sin x} log(t^4 + 1)\, dt.$$

(15 marks)

**<u>Marking Guide:</u>**

Let $F(x) = \int_{\cos x}^{\sin x} log(t^4 + 1)\, dt$. Then,

$$F(x) = \int_{\cos x}^{0} log(t^4 + 1) + \int_{0}^{\sin x} log(t^4 + 1)\, dt$$
$$= -\int_{0}^{\cos x} log(t^4 + 1)\, dt + \int_{0}^{\sin x} log(t^4 + 1)\, dt.$$

(4 marks)

Let $u_1(x) = \cos x$ and $u_2(x) = \sin x$. Then, $\frac{du_1}{dx} = -\sin x$ and $\frac{du_2}{dx} = \cos x$.

(3 marks)

By the chain rule of differentiation and the Fundamental Theorem of Calculus, Part 1,

$$F'(x) = -\frac{d}{du_1}\int_0^{u_1} log(t^4+1)\,dt \cdot \frac{du_1}{dx} + \frac{d}{du_2}\int_0^{u_2} log(t^4+1)\,dt$$

$$= -\,log(u_1^4+1)\cdot(-\,sin\,x) + log(u_2^4+1)\cdot cos\,x$$
$$= sin\,x\,log(cos^4\,x+1) + cos\,x\,log(cos^4\,x+1).$$

(8 marks)
(Total: 15 marks)

**Question 4**

Employ various techniques of integration to evaluate the following integrals.

(a) $\int \frac{x^4+x^3+x^2-x+1}{(x-1)(x^2-3x+3)}\,dx$

(6 marks)

(b) $\int \sqrt{\frac{x-9}{x}}\,dx$

(7 marks)

(c) $\int_1^{\sqrt{3}} \frac{tan^{-1}x}{x^2}\,dx$

(7 marks)

**<u>Marking Guide:</u>**

(a) We first perform long division on the integrand; rewriting the denominator as $x^3-4x^2$ +6x-3, we get

$$\frac{x^4+x^3+x^2-x+1}{(x-1)(x^2-3x+3)} = x+5+\frac{15x^2+28x+16}{(x-1)(x^2-3x+3)}.$$

Next, we resolve the last term into partial fractions. Let $A, B, C$ be constants such that

$$\frac{15x^2-28x+16}{(x-1)(x^2-3x+3)} = \frac{A}{x-1}+\frac{Bx+C}{x^2-3x+3}.$$

(Observe that $x^2-3x+3$ is an irreducible factor.) By the cover-up rule,

$$A = \frac{15(1^2)-28(1)+16}{1^2-3(1)+3} = 3.$$

Moving the partial fraction involving $A$ to the left-hand side of the equation,

$$\frac{15x^2-28x+16}{(x-1)(x^2-3x+3)}-\frac{3}{x-1}=\frac{15x^2-28x+16-3x^2+9x-9}{(x-1)(x^2-3x+3)}$$
$$=\frac{12x^2-19x+7}{(x-1)(x^2-3x+3)}=\frac{12x-7}{x^2-3x+3}=\frac{Bx+C}{x^2-3x+3}.$$

Comparing coefficients of powers of $x$ gives $B=12$ and $C=-7$.

(2 marks)

Therefore

$$\int\frac{x^4+x^3+x^2-x+1}{(x-1)(x^2-3x+3)}dx=\int\left(x+5+\frac{3}{x-1}+\frac{12x-7}{x^2-3x+3}\right)dx$$
$$=\frac{x^2}{2}+5x+3\,log(x-1)+6\int\frac{2x-3}{x^2-3x+3}dx+11\int\frac{1}{\left(x-\frac{3}{2}\right)^2+\frac{3}{4}}dx$$
$$=\frac{x^2}{2}+5x+3\,log(x-1)+6\,log(x^2-3x+3)+11$$
$$\cdot\sqrt{\frac{4}{3}}tan^{-1}\left(\frac{x-\frac{3}{2}}{\sqrt{\frac{3}{4}}}\right)+C$$
$$=\frac{x^2}{2}+5x+3\,log(x-1)+6\,log(x^2-3x+3)+\frac{22}{\sqrt{3}}tan^{-1}\left(\frac{2x-3}{\sqrt{3}}\right)$$
$$+C.$$

(4 marks)
(Total: 6 marks)

(b) Observe that the integrand is defined if and only if $x\geq 9$. Let $x=u^2$ and $dx=2udu$. Then,

$$\int\sqrt{\frac{x-9}{x}}dx=\int\sqrt{\frac{u^2-9}{u^2}}\cdot 2udu=2\int\sqrt{u^2-9}du.$$

(1 mark)

Now make the further substitutions $u = 3\,sec\,\theta$ and $du = 3\,sec\,\theta\,tan\,\theta\,d\theta$. Then the integral becomes

$$2\int \sqrt{9\,sec^2\,\theta - 9}\cdot 3\,sec\,\theta\,tan\,\theta\,d\theta = 18\int\ sec\,\{\theta\}\,tan^2\,\theta\,d\theta$$
$$= 18\int (sec^3\,\theta - sec\,\theta)\,d\theta = 18\left[\int sec^3\,\theta\,d\theta - log|sec\,\theta + tan\,\theta|\right].$$

(1 mark)

To evaluate $\int sec^3\,\theta\,d\theta$, we use the integration-by-parts formula, setting $u_1 = sec\,\theta$ and $dv_1 = sec^2\,\theta\,d\theta$, giving $du_1 = sec\,\theta\,tan\,\theta\,d\theta$ and $v_1 = tan\,\theta$. Then,

$$\int sec^3\,\theta\,d\theta = sec\,\theta\,tan\,\theta - \int tan^2\,\theta\,sec\,\theta\,d\theta$$
$$= sec\,\theta\,tan\,\theta - \int sec^3\,\theta\,d\theta + \int sec\,\theta\,d\theta$$
$$= sec\,\theta\,tan\,\theta + log|sec\,\theta + tan\,\theta| - \int sec^3\,\theta\,d\theta,$$

where we used the identity $tan^2\,\theta = sec^2\,\theta - 1$. Moving the term $\int sec^3\,\theta\,d\theta$ to the left-hand side of the equation and dividing by 2 gives

$$\int sec^3\,\theta\,d\theta = \frac{1}{2}(sec\,\theta\,tan\,\theta + log|sec\,\theta + tan\,\theta|) + C.$$

(2 marks)

Thus,

$$\int \sqrt{\frac{x-9}{x}}\,dx = 18\left[\frac{1}{2}(sec\,\theta\,tan\,\theta + log|sec\,\theta + tan\,\theta|) - log|sec\,\theta + tan\,\theta|\right] + C$$
$$= 9(sec\,\theta\,tan\,\theta - log|sec\,\theta + tan\,\theta|) + C.$$

One has $sec\,\theta = \frac{u}{3} = \frac{\sqrt{x}}{3}$ and so $tan\,\theta = \frac{\sqrt{u^2-9}}{3} = \frac{\sqrt{x-9}}{3}$, giving

$$\int \sqrt{\frac{x-9}{x}}\,dx = 9\left(\frac{\sqrt{x(x-9)}}{9} - log\left|\frac{\sqrt{x}+\sqrt{x-9}}{3}\right|\right) + C$$
$$= \sqrt{x(x-9)} - 9\,log\,\frac{\sqrt{x}+\sqrt{x-9}}{3} + C.$$

(3 marks)
(Total: 7 marks)

(c) We apply the integration-by-parts formula, setting $u = tan^{-1} x$ and $dv = \frac{1}{x^2} dx$, which give $du = \frac{1}{1+x^2} dx$ and $v = -\frac{1}{x}$.

(2 marks)

Then,

$$\int_1^{\sqrt{3}} \frac{tan^{-1} x}{x^2} dx = \left[-\frac{tan^{-1} x}{x}\right]_1^{\sqrt{3}} - \int_1^{\sqrt{3}} -\frac{1}{x(x^2+1)} dx$$
$$= -\frac{\pi}{3\sqrt{3}} + \frac{\pi}{4} + \int_1^{\sqrt{3}} \left(\frac{1}{x} + \frac{Ax+B}{x^2+1}\right) dx$$

for some constants $A$ and $B$. One has

$$\frac{1}{x(x^2+1)} - \frac{1}{x} = \frac{1-x^2-1}{x(x^2+1)} = \frac{-x}{x^2+1} = \frac{Ax+B}{x^2+1},$$

so $A = -1$ and $B = 0$.

(2 marks)

Thus,

$$\int_1^{\sqrt{3}} \frac{tan^{-1} x}{x^2} dx = \pi\left(\frac{1}{4} - \frac{1}{3\sqrt{3}}\right) + [log\, x]_1^{\sqrt{3}} - \left[\frac{1}{2} log(x^2+1)\right]_1^{\sqrt{3}}$$
$$= \pi\left(\frac{1}{4} - \frac{1}{3\sqrt{3}}\right) + log\sqrt{3} - \frac{1}{2} log\, 4 + \frac{1}{2} log\, 2$$
$$= \frac{\pi}{36}\left(9 - 4\sqrt{3}\right) + \frac{1}{2} log\, 3 - 2 \cdot \frac{1}{2} log\, 2 + \frac{1}{2} log\, 2$$
$$= \frac{\pi}{36}\left(9 - 4\sqrt{3}\right) + \frac{1}{2} log\left(\frac{3}{2}\right).$$

(3 marks)
(Total: 7 marks)

**Question 5**

(a) Calculate the area between the curves $y = \frac{1}{x}$, $y^2 = x^3$ and $y = 27$.

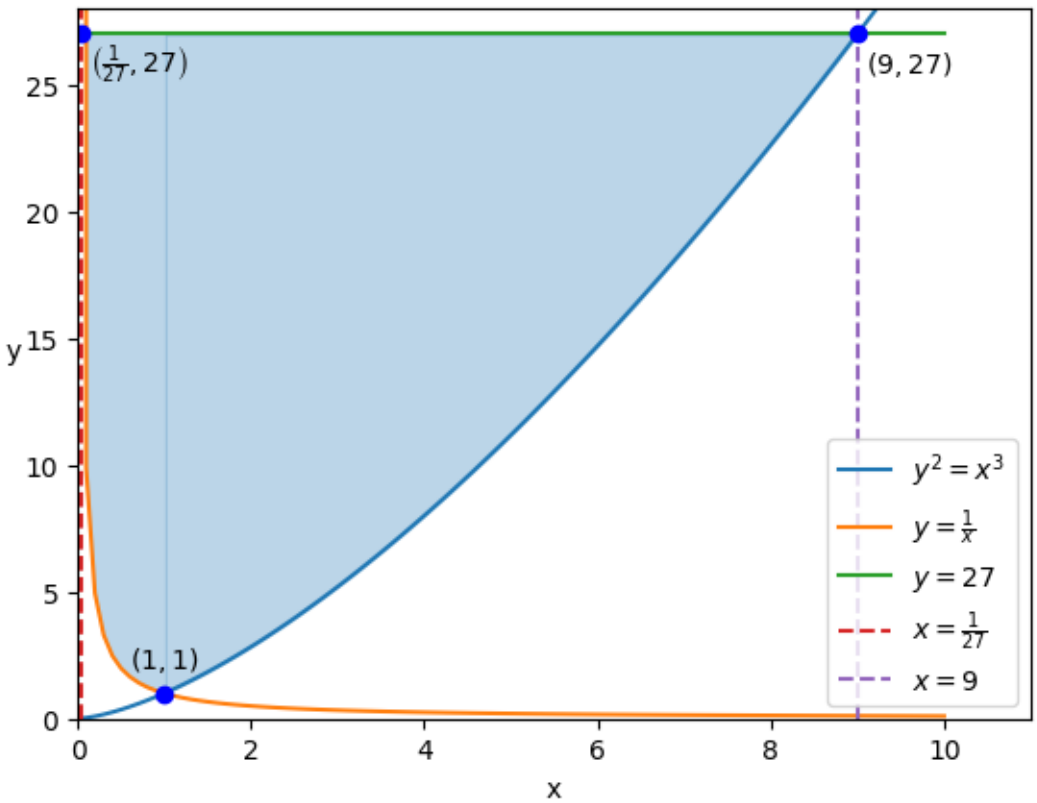
(7 marks)

(b) Calculate the volume of the solid generated by revolving the area between the curves $x^2 + \frac{y^2}{16} = 1$ and $y^2 = 24x$ about the $x$-axis.

(8 marks)

**Marking Guide:**

(a) The region of interest is shown in the next figure.




The point of intersection of the curves $y = \frac{1}{x}$ and $y = 27$ is $\left(\frac{1}{27}, 27\right)$ and the point of intersection of the curves $y^2 = x^3$ and $y = 27$ is $(9,27)$. The $x$-coordinate of the point of intersection of the curves $y^2 = x^3$ and $y = \frac{1}{x}$ is given by the solution of the equation

$$x^{\frac{3}{2}} = \frac{1}{x},$$
$$x^{\frac{5}{2}} = 1,$$

or $x = 1$. Hence the point of intersection of the curves $y^2 = x^3$ and $y = \frac{1}{x}$ is $(1,1)$.

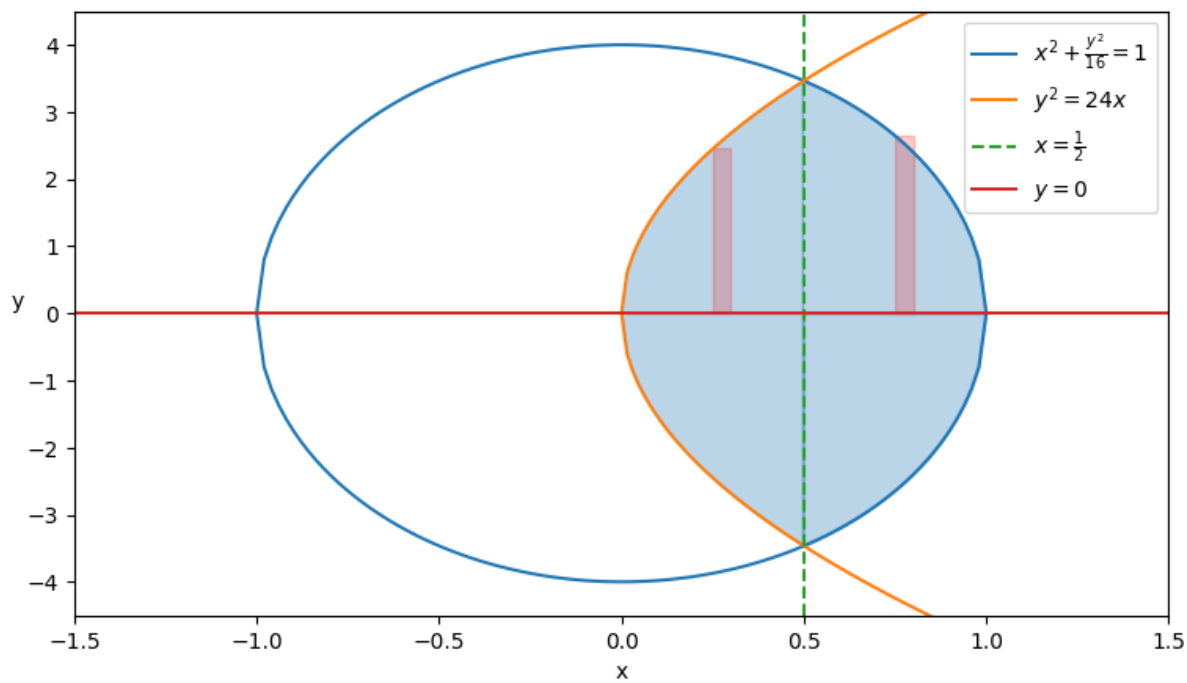
(2 marks)

To find the area of the shaded region, we integrate over the $y$-axis from $y = 1$ to $y = 27$. The region is bounded above by the curve $x = y^{\frac{2}{3}}$ and bounded below by the curve $x = \frac{1}{y}$, so the area is given by

$$A = \int_1^{27} \left(y^{\frac{2}{3}} - \frac{1}{y}\right) dy = \left[\frac{3}{5} y^{\frac{5}{3}} - log\, y\right]_1^{27} = \frac{729}{5} - log\, 27 - \frac{3}{5} + 0 = \frac{726}{5} - log\, 27.$$

(5 marks)
(Total: 7 marks)

(b) The region of interest is shown in the next figure.




We split the area into two parts, the first from $x = 0$ to $x = \frac{1}{2}$ (call this area $A_1$), and the second from $x = \frac{1}{2}$ to $x = 1$ (call this area $A_2$). The $x$-coordinate of the two points of intersection of the curves $x^2 + \frac{y^2}{16} = 1$ and $y^2 = 24x$ is given by the positive solution of the equation

$$x^2 + \frac{24x}{16} = 1,$$
$$16x^2 + 24x - 16 = 0,$$
$$2x^2 + 3x - 2 = 0,$$
$$(2x - 1)(x + 2) = 0.$$

Hence the $x$-coordinate of the two points of intersection is $\frac{1}{2}$.

(2 marks)

Using the disc method, we calculate the volume of the solid generated by summing the volume generated by revolving $A_1$ about the $x$-axis and the volume generated by revolving $A_2$ about the $x$-axis. (The cross-sections of typical half-discs for $A_1$ and $A_2$ are shown as red strips in the figure.) The volume of the solid generated is thus given by

$$V = \pi \int_0^{\frac{1}{2}} 24x dx + \pi \int_{\frac{1}{2}}^{1} \left(\sqrt{16(1-x^2)}\right)^2 dx = \pi \left( \left[\frac{24}{2}x^2\right]_0^{\frac{1}{2}} + 16\left[x - \frac{x^3}{3}\right]_{\frac{1}{2}}^{1} \right)$$
$$= \pi \left(3 + 16\left(1 - \frac{1}{3} - \frac{1}{2} + \frac{1}{24}\right)\right) = \frac{19}{3}\pi.$$

(6 marks)

(Total: 8 marks)

**----- END OF MARKING GUIDE -----**